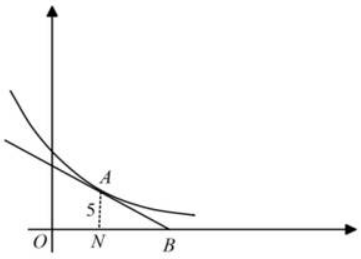
Type MS · Finding the equation of a tangent

Answer 1 of 9 · Question 8 part b · 5 marks

2020 NOV · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

8(a)	$5e^{-2x}+4=e^{2x}$ $\quad 5e^{-2x}+4-e^{2x}=0$ OR	$y=\frac{5}{y}+4\Rightarrow y^2-4y-5=0$	M1
	$(5e^{-x}-e^{x})(e^{-x}+e^{x})=0$	$(y-5)(y+1)=0$	M1
	$5e^{-x}=e^{x}$ $\quad e^{2x}=5$ $\quad x=\frac{1}{2}\ln 5$ $\left(\text{oe eg } \ln\sqrt{5}\right)$	$y=5$	A1
	$\left(e^{-x}=-e^{x} \text{ not possible}\right)$	$e^{2x}=5$ $\quad x=\frac{1}{2}\ln 5$	
	A is $\left(\frac{1}{2}\ln 5, 5\right)$		A1 (4)
(b)	$y=5e^{-2x}+4\Rightarrow \frac{dy}{dx}=-10e^{-2x}$		M1
	At A $\quad \frac{dy}{dx}=-10e^{-2x}=-10\times\frac{1}{5}=-2$		A1ft
	Eqn tgt: $\quad y-5=-2\left(x-\frac{1}{2}\ln 5\right)$		dM1A1
	$y=0\Rightarrow x=\frac{1}{2}(5+\ln 5)$ $\quad (=x \text{ coordinate of } B)^*$		A1cso (5)
ALT	For last 3 marks:		
	Hence $\frac{5}{NB}=2\Rightarrow NB=\frac{5}{2}$		dM1A1
	$ON=\frac{1}{2}\ln 5$		
	$OB=\frac{1}{2}\ln 5+\frac{5}{2}=\frac{1}{2}(5+\ln 5)$ * 		A1cso
(c)	$C_2: \frac{dy}{dx}=2e^{2x}\Rightarrow$ grad tgt at A is $2\times 5=10$		B1ft
	Eqn tgt: $\quad y-5=10\left(x-\frac{1}{2}\ln 5\right)$		M1
	At D: $\quad x=\frac{1}{2}(-1+\ln 5)$		A1
	Area $\triangle ABD=\frac{1}{2}\left(\frac{1}{2}(5+\ln 5)-\frac{1}{2}(-1+\ln 5)\right)\times 5$		M1A1
	$=\frac{15}{2}$ or $7\frac{1}{2}$ $\left(\text{units}^2\right)$		A1 (6)
	See notes for area by "determinant" method		

Type MS · Finding the equation of a tangent

Answer 1 of 9 · Question 8 part b · 5 marks

2020 NOV · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

ALT	For second and third marks: $\frac{5}{ND} = 10 \Rightarrow ND = \frac{1}{2}$ $OD = \frac{1}{2}\ln 5 - \frac{1}{2}$	 M1 A1	 [15]

(a)	
M1	Equate the 2 curve equations. No need to simplify
M1	Factorise their equation
A1	Obtain the one possible value for x (other root need not be seen; if seen it must be rejected) **Must be exact**
A1	Obtain the corresponding value for y. **Must be exact.** Need not be shown in coordinate brackets. Use of $e^{2x} = 5$ leads to $y = 5$ without use of a value of x, so M1M1A0A1 can be scored. There must only be one correct y shown. Accept $y = e^{\ln 5}$
(b)	
M1	Differentiate the equation of C_1 $5e^{-2x} \rightarrow ke^{-2x}$ where $k = \pm 5$ or ± 10 and no integration seen
A1ft	Grad at A = -2 follow through their x coordinate
dM1	Obtain the equation of the tangent at A using their gradient and their coordinates of A. Can be in any form but if $y = mx + c$ is used a value for c must be found. Gradient of the tangent must be numerical.
A1	Correct equation in any form
A1cso	Correct x coordinate of B obtained from correct working.
ALT	For last 3 marks
dM1	Use their y coordinate of A and their (numerical) gradient of the tangent to find the length NB (where N is the foot of the perpendicular from A to the x-axis)
A1	Correct length of NB
A1cso	Add the x coordinate of A to obtain the x coordinate of B
(c)	
B1ft	Correct gradient of tangent to C_2 at A follow through their x coordinate
M1	Obtain an equation for the tangent using their gradient and their coordinates of A Gradient of the tangent must be numerical.
A1	Correct x coordinate of D (exact or minimum 3 sf)
M1	Use a correct formula for the area of a triangle with their y coordinate of A, their x coordinate of D and the **given** x coordinate of B
A1	Correct, unsimplified area Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5
	Heron's formula: Nos which may be seen: $AB = \frac{5\sqrt{5}}{2}, AD = \frac{\sqrt{101}}{2}, BD = 3, s = \frac{1}{2}(a+b+c) = 6.8$
ALT	For second and third marks:
M1	Use their y coordinate of A and their gradient of the tangent to find the length ND
A1	Use the x coordinate of A to obtain the x coordinate of D

Type MS · Finding the equation of a tangent

Answer 1 of 9 · Question 8 part b · 5 marks

2020 NOV · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

ALT	Area by "determinant" method:
M1	Eg Area $= \frac{1}{2}\begin{vmatrix} \frac{1}{2}\ln 5 & \frac{1}{2}(5+\ln 5) & \frac{1}{2}(\ln 5-1) & \frac{1}{2}\ln 5 \\ 5 & 0 & 0 & 5 \end{vmatrix}$ y coordinates of B and D must be 0 Must include the ½ and have 4 sets of coordinates with first and last the same.
A1	$= \frac{1}{2}\left(\frac{1}{2}(\ln 5-1)\times 5 - \frac{1}{2}(\ln 5+1)\right)$ Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5 **Must be positive**

Type MS · Finding the equation of a tangent

Answer 1 of 9 · Question 8 part b · 5 marks

2020 NOV · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question Number	Scheme	Marks
	$2+4x-x^2$	

Type MS · Finding the equation of a tangent

Answer 2 of 9 · Question 6 part b · 6 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 6 · find part b in this mark scheme extract

6 (a)	$x^2-5x+4=(x-4)(x-1)$	M1
	$A(1,0)$ $B(4,0)$	A1 A1 (3)
	Notes	
(a)		
M1	For solving the quadratic	
A1	For $A(1,0)$	
A1	For $B(4,0)$	
(b)	$\frac{dy}{dx}=2x-5$	M1
	When $x=1$ $\frac{dy}{dx}=-3$	A1
	Tangents meet on the axis of symmetry of curve C so $x=\frac{1+4}{2}=\frac{5}{2}$	M1 A1
	When $x=\frac{5}{2}$ $y=-3\left(\frac{5}{2}-1\right)=-\frac{9}{2}$	M1
	$\left(\frac{5}{2},-\frac{9}{2}\right)$	A1 (6)
	Notes	
(b)		
M1	For $\frac{dy}{dx}=2x-5$ (Allow if seen in part(c))	
A1	$\frac{dy}{dx}=-3$ when $x=1$	
M1	For use of $\frac{x_1+x_2}{2}$ or $\frac{-b}{2a}$ or $\frac{dy}{dx}=0$	
A1	$x=\frac{5}{2}$	
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m(x-x_1)$ oe	
A1	For $\left(\frac{5}{2},-\frac{9}{2}\right)$	

Type MS · Finding the equation of a tangent

Answer 2 of 9 · Question 6 part b · 6 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 6 · find part b in this mark scheme extract

	Alternative Method $\frac{dy}{dx} = 2x - 5$ When $x = 1$ $\frac{dy}{dx} = -3$ So $y = -3x + 3$ When $x = 4$ $\frac{dy}{dx} = 3$ So $y = 3x - 12$ Meet when $-3x + 3 = 3x - 12$ $\left(\frac{5}{2}, -\frac{9}{2}\right)$	M1 A1 M1 A1 M1 A1 (6)
	Notes	
	Alternative	
M1	For an attempt to find the equation of the tangent of C at A	
A1	For $y = -3x + 3$	
M1	For an attempt to find the equation of the tangent of C at B	
A1	For $y = 3x - 12$	
M1	For equating the two equations	
A1	For $\left(\frac{5}{2}, -\frac{9}{2}\right)$	

Type MS · Finding the equation of a tangent

Answer 2 of 9 · Question 6 part b · 6 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 6 · find part b in this mark scheme extract

(c)	Gradient of normal at (0, 1) = $\frac{1}{3}$	M1
	When $x=\frac{5}{2}$ $y=\frac{1}{3}\left(\frac{5}{2}-1\right)=\frac{1}{2}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)
	Alternative Method $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$	M1
	Meet when $\frac{1}{3}x-\frac{1}{3}=-\frac{1}{3}x+\frac{4}{3}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)

	Notes
(c)	
M1	For Gradient of normal at (0, 1) = $\frac{1}{3}$ ft the gradient found in part (b)
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m\left(x-x_1\right)$ oe
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$
	Alternative
M1	For $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$
M1	For equating the two equations
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$

Type MS · Finding the equation of a tangent

Answer 2 of 9 · Question 6 part b · 6 marks

2020 NOV · Paper 02 (regional R) · extract 4 of 6 · find part b in this mark scheme extract

(d)	Area $= \frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 1 Area $= \frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 2 $AN=\sqrt{1.5^2+0.5^2}=\frac{\sqrt{10}}{2}$ $AT=\sqrt{1.5^2+4.5^2}=\frac{3\sqrt{10}}{2}$	M1
	Area $= 2\times\frac{1}{2}\times\frac{\sqrt{10}}{2}\times\frac{3\sqrt{10}}{2}=\frac{30}{4}=\frac{15}{2}$	M1 A1 (3)
	Alternative Method 3 Area $= \frac{1}{2}\begin{vmatrix} 1 & \frac{5}{2} & 4 & \frac{5}{2} & 1 \\ 0 & -\frac{9}{2} & 0 & \frac{1}{2} & 0 \end{vmatrix} \Rightarrow \frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
		[15]

	Notes
(d)	
M1	For $\frac{1}{2}\times AB\times NT$
M1	For $\frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)$
A1	For $\frac{15}{2}$
	Alternative 1
M1	For area of triangle ANB + area of triangle ATB
M1	For $\frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}$
A1	$\frac{15}{2}$
	Alternative 2
M1	For finding AN and AT
M1	For $2\times\frac{1}{2}\times AN\times AT$
A1	$\frac{15}{2}$

Type MS · Finding the equation of a tangent

Answer 2 of 9 · Question 6 part b · 6 marks

2020 NOV · Paper 02 (regional R) · extract 5 of 6 · find part b in this mark scheme extract

	Alternative 3
M1	Use of area $= \frac{1}{2}\begin{vmatrix} 1 & a & 4 & c & 1 \\ 0 & b & 0 & d & 0 \end{vmatrix}$ oe where (a, b) and (c, d) are the coordinates of T and N
M1	$\frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)$
A1	$\frac{15}{2}$

Type MS · Finding the equation of a tangent

Answer 2 of 9 · Question 6 part b · 6 marks

2020 NOV · Paper 02 (regional R) · extract 6 of 6 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a tangent

Answer 3 of 9 · Question 6 part a · 5 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

Number		
6 (a)	$\frac{dy}{dx} = \frac{2}{\sqrt{x}}$	M1
	When $x = 9$ $\frac{dy}{dx} = \frac{2}{3}$	A1
	$y - 12 = \frac{2}{3}(x - 9)$	M1
	When $y = 0$ $-12 = \frac{2}{3}(x - 9)$	M1
	$x = -9$ So $(-9, 0)$	A1 (5)
(b)	Gradient of Normal: $-\frac{3}{2}$	M1
	$y - 12 = -\frac{3}{2}(x - 9)$	M1
	When $y = 0$ $-12 = -\frac{3}{2}(x - 9)$	M1
	$x = 17$ So $(17, 0)$	A1 (4)
(c)	$\frac{1}{2} \times 12 \times 26 = 156$	M1 A1 (2)
		Total 11 marks

Type MS · Finding the equation of a tangent

Answer 3 of 9 · Question 6 part a · 5 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt to differentiate y wrt x. Accept $\frac{dy}{dx} = 2x^{-\frac{1}{2}}$ Allow as a minimum $\frac{dy}{dx} = kx^{-\frac{1}{2}}$ where k is a constant $\neq 0$
	A1	For $\frac{dy}{dx} = \frac{2}{3}$
	M1	For a complete method to find the equation of a straight line. If they use $y = mx + c$ then they must reach a value for c for this mark. Accept the equation of the line with their value of $\frac{dy}{dx}$ using the given coordinates.
	M1	For substitution of $y = 0$ to find a value for x
	A1	For $(-9, 0)$
(b)	M1	For gradient of Normal: $-\frac{3}{2}$ Which is the negative reciprocal of their gradient obtained in (a)
	M1	For a complete method to find the equation of a straight line with a gradient of $-\frac{3}{2}$ (ft their gradient of normal) If they use $y = mx + c$ then they must reach a value for c for this mark.
	M1	For substitution of $y = 0$ to find a value for x
	A1	For $(17, 0)$
(c)	M1	For any correct method for finding the area of a triangle. e.g., $A = \frac{1}{2} \times 12 \times ('17' - '-9') = 156$ OR $A = \frac{1}{2}\begin{bmatrix} 9 & -9 & 17 & 9 \\ 12 & 0 & 0 & 12 \end{bmatrix} = \frac{1}{2}\left[(0+0+204)-(0+0-108)\right] = 156$
	A1	For 156

Type MS · Finding the equation of a tangent

Answer 3 of 9 · Question 6 part a · 5 marks

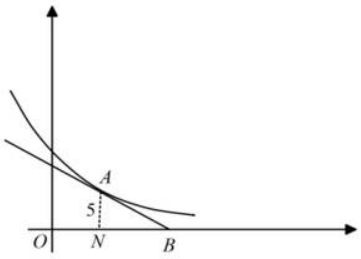
2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a tangent

Answer 4 of 9 · Question 8 part c · 6 marks

2020 NOV · Paper 02 · extract 1 of 4 · find part c in this mark scheme extract

8(a)	$5e^{-2x}+4=e^{2x}$ $\quad 5e^{-2x}+4-e^{2x}=0$ OR	$y=\frac{5}{y}+4\Rightarrow y^2-4y-5=0$	M1
	$(5e^{-x}-e^{x})(e^{-x}+e^{x})=0$	$(y-5)(y+1)=0$	M1
	$5e^{-x}=e^{x}$ $\quad e^{2x}=5$ $\quad x=\frac{1}{2}\ln 5$ $\left(\text{oe eg } \ln\sqrt{5}\right)$	$y=5$	A1
	$\left(e^{-x}=-e^{x} \text{ not possible}\right)$	$e^{2x}=5$ $\quad x=\frac{1}{2}\ln 5$	
	A is $\left(\frac{1}{2}\ln 5, 5\right)$		A1 (4)
(b)	$y=5e^{-2x}+4\Rightarrow \frac{dy}{dx}=-10e^{-2x}$		M1
	At A $\quad \frac{dy}{dx}=-10e^{-2x}=-10\times\frac{1}{5}=-2$		A1ft
	Eqn tgt: $\quad y-5=-2\left(x-\frac{1}{2}\ln 5\right)$		dM1A1
	$y=0\Rightarrow x=\frac{1}{2}(5+\ln 5)$ $\quad (=x \text{ coordinate of } B)^*$		A1cso (5)
ALT	For last 3 marks:		
	Hence $\frac{5}{NB}=2\Rightarrow NB=\frac{5}{2}$		dM1A1
	$ON=\frac{1}{2}\ln 5$		
	$OB=\frac{1}{2}\ln 5+\frac{5}{2}=\frac{1}{2}\ 5+\ln 5$ *		A1cso
(c)	$C_2: \frac{dy}{dx}=2e^{2x}\Rightarrow$ grad tgt at A is $2\times 5=10$		B1ft
	Eqn tgt: $\quad y-5=10\left(x-\frac{1}{2}\ln 5\right)$		M1
	At D: $\quad x=\frac{1}{2}(-1+\ln 5)$		A1
	Area $\triangle ABD=\frac{1}{2}\left(\frac{1}{2}(5+\ln 5)-\frac{1}{2}(-1+\ln 5)\right)\times 5$		M1A1
	$=\frac{15}{2}$ or $7\frac{1}{2}$ $\left(\text{units}^2\right)$		A1 (6)
	See notes for area by "determinant" method		

Type MS · Finding the equation of a tangent

Answer 4 of 9 · Question 8 part c · 6 marks

2020 NOV · Paper 02 · extract 2 of 4 · find part c in this mark scheme extract

ALT	For second and third marks: $\frac{5}{ND} = 10 \Rightarrow ND = \frac{1}{2}$ $OD = \frac{1}{2}\ln 5 - \frac{1}{2}$	 M1 A1 [15]

(a)	
M1	Equate the 2 curve equations. No need to simplify
M1	Factorise their equation
A1	Obtain the one possible value for x (other root need not be seen; if seen it must be rejected) **Must be exact**
A1	Obtain the corresponding value for y. **Must be exact.** Need not be shown in coordinate brackets. Use of $e^{2x} = 5$ leads to $y = 5$ without use of a value of x, so M1M1A0A1 can be scored. There must only be one correct y shown. Accept $y = e^{\ln 5}$
(b)	
M1	Differentiate the equation of C_1 $5e^{-2x} \rightarrow ke^{-2x}$ where $k = \pm 5$ or ± 10 and no integration seen
A1ft	Grad at A = -2 follow through their x coordinate
dM1	Obtain the equation of the tangent at A using their gradient and their coordinates of A. Can be in any form but if $y = mx + c$ is used a value for c must be found. Gradient of the tangent must be numerical.
A1	Correct equation in any form
A1cso	Correct x coordinate of B obtained from correct working.
ALT	For last 3 marks
dM1	Use their y coordinate of A and their (numerical) gradient of the tangent to find the length NB (where N is the foot of the perpendicular from A to the x-axis)
A1	Correct length of NB
A1cso	Add the x coordinate of A to obtain the x coordinate of B
(c)	
B1ft	Correct gradient of tangent to C_2 at A follow through their x coordinate
M1	Obtain an equation for the tangent using their gradient and their coordinates of A Gradient of the tangent must be numerical.
A1	Correct x coordinate of D (exact or minimum 3 sf)
M1	Use a correct formula for the area of a triangle with their y coordinate of A, their x coordinate of D and the **given** x coordinate of B
A1	Correct, unsimplified area Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5
	Heron's formula: Nos which may be seen: $AB = \frac{5\sqrt{5}}{2}, AD = \frac{\sqrt{101}}{2}, BD = 3, s = \frac{1}{2}(a+b+c) = 6.8$
ALT	For second and third marks:
M1	Use their y coordinate of A and their gradient of the tangent to find the length ND
A1	Use the x coordinate of A to obtain the x coordinate of D

Type MS · Finding the equation of a tangent

Answer 4 of 9 · Question 8 part c · 6 marks

2020 NOV · Paper 02 · extract 3 of 4 · find part c in this mark scheme extract

ALT	Area by "determinant" method:
M1	Eg Area $= \frac{1}{2}\begin{vmatrix} \frac{1}{2}\ln 5 & \frac{1}{2}(5+\ln 5) & \frac{1}{2}(\ln 5-1) & \frac{1}{2}\ln 5 \\ 5 & 0 & 0 & 5 \end{vmatrix}$ y coordinates of B and D must be 0 Must include the ½ and have 4 sets of coordinates with first and last the same.
A1	$= \frac{1}{2}\left(\frac{1}{2}(\ln 5-1)\times 5 - \frac{1}{2}(\ln 5+1)\right)$ Allow use of correct but non-exact coordinates
A1	Correct area Accept only $7\frac{1}{2}, \frac{15}{2}$ or 7.5 **Must be positive**

Type MS · Finding the equation of a tangent

Answer 4 of 9 · Question 8 part c · 6 marks

2020 NOV · Paper 02 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	$2+4x-x^2$	

Type MS · Finding the equation of a tangent

Answer 5 of 9 · Question 4 part whole · 7 marks

2023 JAN · Paper 02 · find part whole in this mark scheme extract

4	When $x = \frac{\pi}{2}$ $\frac{dy}{dx} = 3\left(\frac{\pi}{2}\right)^{\tilde{}} \sin\left(\frac{\pi}{2}\right) + \left(\frac{\pi}{2}\right)^{\tilde{}} \cos\left(\frac{\pi}{2}\right) = \left[\text{'}\frac{3\pi^{\tilde{}}}{4}\text{'}\right]$	M1
	$y - \frac{\pi^3}{8} = \frac{3\pi^2}{4}\left(x - \frac{\pi}{2}\right)$	M1
	$y = \frac{3}{4}\pi^2 x - \frac{1}{4}\pi^3$	A1
		Total 7 marks

Type MS · Finding the equation of a tangent

Answer 6 of 9 · Question 3 part b · 3 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

3(a)	$\frac{dy}{dx} = 2e^{2x}\left(x^2+1\right)+e^{2x}\left(2x\right)$	M1A1A1 [3]
(b)	When $x = 0$ $\frac{dy}{dx} = 2\times1\times1+1\times0=2$ $\quad y = e^{2\times0}\left(0+1\right)=1$ $y-1=2\left(x-0\right)\Rightarrow y=2x+1$	B1B1 B1 [3]
		Total 6 marks

Type MS · Finding the equation of a tangent

Answer 6 of 9 · Question 3 part b · 3 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks
2 (a)	$7+4x-x^2=11-(x-2)^2$ $[a=11, b=1, c=-2]$ **ALT**	M1A1A1 [3]

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

2025 NOV · Paper 02 · extract 1 of 10 · find part a in this mark scheme extract

10(a)	At A:	
	$\left[y = 3(4-0)^{\frac{1}{2}} = 3\times 2 = \right] 6$	B1
	$\left[\frac{dy}{dx} = \right] 3\times\frac{1}{2}\times -1\times(4-x)^{-\frac{1}{2}} \left[= -\frac{3}{2}(4-x)^{-\frac{1}{2}}\right]$ oe	M1
	When $x = 0$:	
	$\left[\frac{dy}{dx} = -\frac{3}{2}(4-0)^{-\frac{1}{2}} = \right] -\frac{3}{4}$ oe	A1
	Equation of l:	
	eg $y-6 = -\frac{3}{4}(x-0)$ or $y = -\frac{3}{4}x+6$ or $4y = -3x+24$ oe	M1A1 [5]
(b)	When $y = 0$:	
	$\left[\text{Curve: } 0 = y = 3(4-x)^{\frac{1}{2}} \Rightarrow x = \right] 4$	B1ft
	$[\text{Line: } 4\times 0+3x-24 = 0 \Rightarrow x =] 8$	
	$[V =]\pi\int_0^{"8"}\left("-\frac{3}{4}x+6"\right)^2 dx - \pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$ or	M1
	$[V =]\frac{1}{3}\times\pi\times 6^2\times"8"-\pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$	
	$\left[= \pi\int_0^{"8"}\left(\frac{9}{16}x^2-9x+36\right)dx - \pi\int_0^{"4"}(36-9x)\,dx\right]$	
	$= [\pi]\left[\frac{3}{16}x^3-\frac{9}{2}x^2+36x\right]_{[0]}^{["8"]} - [\pi]\left[36x-\frac{9x^2}{2}\right]_{[0]}^{["4"]}$	M1M1 A1ft
	$\left[= \pi\left[\frac{3}{16}\times"8"^3-\frac{9}{2}\times"8"^2+36\times"8"\right]-\pi\left[36\times"4"-\frac{9\times"4"^2}{2}\right]\right.$	
	$\left.[= 96\pi-72\pi =]24\pi\right]$	[5]
		Total 10 marks

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

2025 NOV · Paper 02 · extract 2 of 10 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct unsimplified vector in **a** and **b** for $\overrightarrow{BC}$ or for $\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$
	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BC}$
	M1	For $\overrightarrow{BC} = \frac{1}{2}\overrightarrow{AD}$ oe (note accept eg $2BC = AD$ ie without vector notation for this mark) Accept equivalences – see example MS.
	A1	For the correct conclusion, with all previous marks gained. Ignore work on other vectors. Do not accept eg $2BC = AD$ without vector notation for this mark Note B0 B1 M1 A0 is a potential score. $2BC = AD$ followed by eg $10\mathbf{a} + 6\mathbf{b} = 10\mathbf{a} + 6\mathbf{b}$ will get Aq
Watch for equivalences such as working out vector $\overrightarrow{CB}$ and stating $\overrightarrow{CB} = -\frac{1}{2}\overrightarrow{AD}$ Permit eg $\frac{\overrightarrow{AD}}{\overrightarrow{BC}} = 2$		
(b)	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BD}$
	B1ft	For applying Pythagoras theorem fully and correctly to find the length BD Ft any vector believed to be their $\overrightarrow{BD}$. $8^2 + 2^2$ alone is insufficient.
	M1	For using their vector for $\overrightarrow{BD}$ with their length BD to attempt to form a unit vector
	A1	For the correct unit vector.

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

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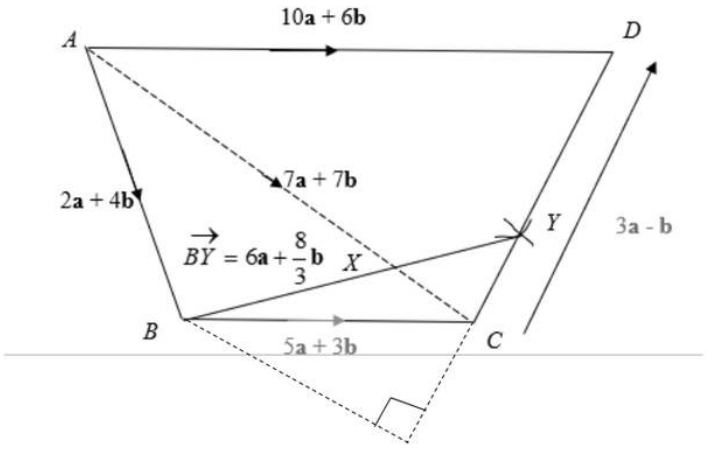
(c)(i)	$\left[\left[\overrightarrow{CY}=\right]\frac{1}{3}(3\mathbf{a}-\mathbf{b})=\mathbf{a}-\frac{1}{3}\mathbf{b}$ oe eg $\overrightarrow{YC}$ or $\left[\left[\overrightarrow{YD}=\right]\frac{2}{3}(3\mathbf{a}-\mathbf{b})=2\mathbf{a}-\frac{2}{3}\mathbf{b}$ oe eg $\overrightarrow{DY}$ Note: $\overrightarrow{BC}=5\mathbf{a}+4\mathbf{b}$ $\left[\overrightarrow{BY}=5\mathbf{a}+3\mathbf{b}+\mathbf{a}-\frac{\mathbf{b}}{3}=\right]6\mathbf{a}+\frac{8}{3}\mathbf{b}\Rightarrow$	M1
	$\left[\overrightarrow{BX}=\right]\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$	dM1
	$\left[\overrightarrow{BX}=\right]-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $\left[\overrightarrow{BX}=\right](5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})$ $\left[\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $(5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})\right]$ $\left[\begin{array}{l}6\lambda=7\mu-2 \text{ and } \frac{8}{3}\lambda=7\mu-4 \text{ or}\\ \frac{8}{3}\lambda=-7\alpha+3 \text{ and } 6\lambda=-7\alpha+5\end{array}\right]$	ddM1
	$\left[\lambda=\frac{3}{5};\mu=\frac{4}{5};\alpha=\frac{1}{5}\Rightarrow\overrightarrow{BX}=\right]\frac{18}{5}\mathbf{a}+\frac{8}{5}\mathbf{b}\Rightarrow[BX:XY=]3:2$	A1 [4]
ALT	First mark as main scheme. $\left[\overrightarrow{XY}=\right]\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$	M1
	$\left[\overrightarrow{XY}=\right]\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\left[\overrightarrow{XY}=\right]\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)$ $\left[\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)\right]$	dM1
	$\left[\begin{array}{l}6\beta=7\gamma+1 \text{ and } \frac{8}{3}\beta=7\gamma-\frac{1}{3} \text{ or}\\ \frac{8}{3}\beta=-7\delta+6+\frac{2}{3} \text{ and } 6\beta=-7\delta+10-2\end{array}\right]$	ddM1
	$\left[\beta=\frac{2}{5};\gamma=\frac{1}{5};\delta=\frac{4}{5}\Rightarrow\overrightarrow{XY}=\right]\frac{12}{5}\mathbf{a}+\frac{16}{15}\mathbf{b}[BX:XY=]3:2$	A1 [4]

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

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Part	Mark	Notes
(c)(i) Main and ALT	M1	Correct simplified vector in **a** and **b** for $\pm \overrightarrow{BY}$
	dM1	Correct vector in **a** and **b** with parameter using $\overrightarrow{CY}$ or $\overrightarrow{YC}$ for $\overrightarrow{BX}$ / $\overrightarrow{XB}$ or $\overrightarrow{XY}$ / $\overrightarrow{YX}$ **and** correct vector in **a** and **b** with a different parameter along a second distinct path using $\overrightarrow{AC}$ or $\overrightarrow{CA}$ for the same vector. Dependent on previous method mark.
	ddM1	For the correct simultaneous equations. Any one correct parameter or $\overrightarrow{BX} = \frac{18}{5}\mathbf{a} + \frac{8}{5}\mathbf{b}$ will imply this mark. Dependent on both previous method marks.
	A1	For the required ratio $BX : XY = 3 : 2$
Note, correct method can include the opposite signs in vectors multiplied by a parameter, as the parameter will just be the opposite sign when calculated and a fully correct solution is possible. Eg $\left[\overrightarrow{BX} = \right](5\mathbf{a}+3\mathbf{b}) + \alpha(7\mathbf{a}+7\mathbf{b})$ would just make $\alpha = -\frac{1}{5}$ Vectors questions notoriously have different methods. Correct answer, not from obviously incorrect working, full marks may be awarded **if method is shown.** (This is because the demand of this question says students must use a vector method). Min method must involve finding two correct different distinct relevant paths.		



Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

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(ii)	$\left[\frac{CXY}{BXC}=\frac{2}{3}\ \text{and}\ \frac{BAC}{BXC}=\frac{5}{1}\Rightarrow\right]\ \frac{CXY}{BAC}=\frac{2}{15}$ oe eg $\frac{BAC}{CXY}=\frac{15}{2}$	M1
Eg 1	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{CXY}{ABCD}=\frac{\frac{2}{15}}{3}\left[=\frac{2}{45}\right]$ oe eg $\frac{ABCD}{CXY}=\frac{3}{\frac{2}{15}}\left[=\frac{45}{2}\right]$	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
Eg 2	$\frac{CXY}{BCY}=\frac{2}{5}$ and $\left[\frac{BCY}{BCD}=\frac{1}{3}\right]\ \frac{BCY}{BCA}=\frac{1}{3}$ oe	M1
	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{BCY}{ABCD}=\frac{\frac{1}{3}}{3}\left[=\frac{1}{9}\right]$ oe and $\frac{CXY}{ABCD}=\frac{\frac{2}{5}}{9}\left[=\frac{2}{45}\right]$ oe	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
		Total 16 marks

Part	Mark	Notes
(ii)	M1	For any correct ratio, in any form, stated between a pair of triangles.
	M1	For any correct second ratio, in any form, stated between a different pair of triangles.
	M1	For any correct ratio, in any form between a triangle and the trapezium.
	A1	For the correct ratio only.
Any one of the ratios listed may be implied by a correct answer with no obviously incorrect working leading to this answer. Examples of other correct ratios: $\frac{BCD}{ABCD}=\frac{1}{3}$ $\frac{ABX}{CXY}=\frac{1}{6}$ **To be checked** Not carrying incorrect ratios through.		

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

2025 NOV · Paper 02 · extract 6 of 10 · find part a in this mark scheme extract

Question	Scheme	Marks
10(a)	At A:	
	$\left[y=3(4-0)^{\frac{1}{2}}=3\times 2=\right]6$	B1
	$\left[\frac{dy}{dx}=\right]3\times\frac{1}{2}\times -1\times(4-x)^{-\frac{1}{2}}\left[=-\frac{3}{2}(4-x)^{-\frac{1}{2}}\right]$ oe	M1
	When $x=0$:	
	$\left[\frac{dy}{dx}=-\frac{3}{2}(4-0)^{-\frac{1}{2}}=\right]-\frac{3}{4}$ oe	A1
	Equation of l:	
	eg $y-6=-\frac{3}{4}(x-0)$ or $y=-\frac{3}{4}x+6$ or $4y=-3x+24$ oe	M1A1 [5]
(b)	When $y=0$:	
	$\left[\text{Curve: } 0=y=3(4-x)^{\frac{1}{2}}\Rightarrow x=\right]4$	B1ft
	$[\text{Line: } 4\times 0+3x-24=0\Rightarrow x=]8$	
	$[V=]\pi\int_0^{"8"}\left("-\frac{3}{4}x+6"\right)^2 dx-\pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$ or	M1
	$[V=]\frac{1}{3}\times\pi\times 6^2\times"8"-\pi\int_0^{"4"}\left(3\sqrt{4-x}\right)^2 dx$	
	$\left[=\pi\int_0^{"8"}\left(\frac{9}{16}x^2-9x+36\right)dx-\pi\int_0^{"4"}(36-9x)\,dx\right]$	
	$=[\pi]\left[\frac{3}{16}x^3-\frac{9}{2}x^2+36x\right]_{[0]}^{["8"]}-[\pi]\left[36x-\frac{9x^2}{2}\right]_{[0]}^{["4"]}$	M1M1 A1ft
	$\left[=\pi\left[\frac{3}{16}\times"8"^3-\frac{9}{2}\times"8"^2+36\times"8"\right]-\pi\left[36\times"4"-\frac{9\times"4"^2}{2}\right]\right]$	
	$\left[=96\pi-72\pi=\right]24\pi$	[5]
		Total 10 marks

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

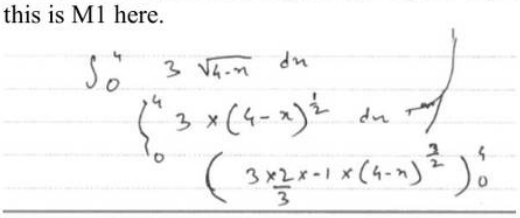
2025 NOV · Paper 02 · extract 7 of 10 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For [$y =$] 6
	M1	For the correct derivative, doesn't need to be simplified.
	A1	For $-\frac{3}{4}$ oe If $-\frac{3}{4}$ oe from incorrect work – send to Review.
	M1	For correctly forming an equation of l using their values for y and $\frac{dy}{dx}$ Not a dependent mark, but there must be some attempt to have substituted $x = 0$ into a changed function to find a gradient. We will also accept if a candidate has done this and then found the negative reciprocal. If using $y = mx + c$, a correct rearrangement must be shown to find c and the equation then fully formed. If the rearrangement is not shown and the final equation is incorrect, c will need to be checked by examiners and must be correct for their values.
	A1	Award this mark for **any correct version** of the equation, simplified or not, rearranged or not, seen at any point. **It does not need to be given in the form stated by the question.**
In a complete break with convention, that must not be applied to any other question on this paper, examiners should: • mark crossed out work, even if replaced • mark multiple attempts (including crossed out work) • award the highest mark from all attempts For the usual conventions on marking crossed out work and multiple attempts, see General Guidance in the Mark Scheme.		

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

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(b)	**For part (b) – allow candidates to follow through $y = mx + c$ selected from (any multiple attempts) in part (a).**	
	B1ft	For $[x =]4$ **and** $[x =]$"8" ft the correct value of x for their $y = mx + c$ when $y = 0$ For example if $y = \frac{3}{4}x + 6$ is used $x = -8$ must be listed.
	M1	For a correct statement for the volume of revolution. Condone $[V =]\pi\int_0^4 \left(3\sqrt{4-x}\right)^2 - \pi\int_0^{"8"} \left("-\frac{3}{4}x+6"\right)^2 \, dx$ or $[V =]\pi\int_0^4 \left(3\sqrt{4-x}\right)^2 - \frac{1}{3}\times\pi\times 6^2\times"8"- \, dx$ This mark will ft their $y = mx + c$
	M1	For a minimally acceptable integration of their expressions, no power of x to decrease. There must be a minimum of 2 terms to integrate. π and limits do not need to be present. Simplification of terms after integration not necessary. If students have chosen to write down and find the volume of a cone, this is for a minimally acceptable integration of their $\left(3\sqrt{4-x}\right)^2$ Sometimes students are forgetting to square expressions. If you see this correct integration, this is M1 here. 
	M1	For a fully correct integration of $\left(3\sqrt{4-x}\right)^2$ π and limits do not need to be present. Simplification of terms not necessary.
	A1ft	For all of $\pi\int_0^{"8"} ("\text{their } mx+c")^2 \, dx - \pi\int_0^4 \left(3\sqrt{4-x}\right)^2 \, dx$ or condone $\pi\int_0^4 \left(3\sqrt{4-x}\right)^2 - \pi\int_0^{"8"} (\text{their } mx+c)^2 \, dx$ seen **and** fully and correctly expanded/integrated – terms do not have to be simplified. If writing down and using the formula for the volume of a cone, for their ft value of x from their equation, this must be evaluated (simplified) correctly as $r\pi \quad r \neq 0$ - examiners to check and there must be a fully correct integration of $\pi\int_0^4 \left(3\sqrt{4-x}\right)^2$. Examiners to check expansion/integration of $(\text{their } mx+c)^2$, π must be present at least once on each integration. We need to see the limits on all integration statements, ft their value of x for the intercept of the line but we do not need to see them substituted unless they are not present eg at the end of square brackets, as is convention. **We do not need to see and are not checking the final answer.**

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

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Note you may also see

$$\pi\int_0^4\left("-\frac{3}{4}x+6"\right)^2\,\mathrm{d}x-\pi\int_0^4\left(3\sqrt{4-x}\right)^2\,\mathrm{d}x\ +\pi\int_4^{"8"}\left("-\frac{3}{4}x+6"\right)^2$$

Type MS · Finding the equation of a tangent

Answer 7 of 9 · Question 10 part a · 5 marks

2025 NOV · Paper 02 · extract 10 of 10 · find part a in this mark scheme extract

Question	Scheme	Marks
Look carefully at all diagrams, including the one provided. **Any contradictions with working, working takes precedence.**		

Type MS · Finding the equation of a tangent

Answer 8 of 9 · Question 10 part c · 4 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

Number		
10 (a)	$\frac{1}{2}+\sin 3x=0 \Rightarrow \sin 3x=-\frac{1}{2} \Rightarrow 3x=-\frac{\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6} \Rightarrow x=\frac{7\pi}{18},\frac{11\pi}{18}$	M1
	Coordinates of M are $\left(\frac{7\pi}{18},\ 0\right)$*	A1
	Coordinates of N are $\left(\frac{11\pi}{18},\ 0\right)$	A1 [3]
	ALT	
	$\frac{1}{2}+\sin\left(3\times\frac{7\pi}{18}\right)=0$	[M1
	Coordinates of M are $\left(\frac{7\pi}{18},0\right)$ *	A1
	Coordinates of N are $\left(\frac{11\pi}{18},0\right)$	A1]
(b)	$\frac{dy}{dx}=3\cos 3x=0 \Rightarrow 3x=\frac{\pi}{2} \Rightarrow x=\frac{\pi}{6}$	M1A1
	$y=\frac{1}{2}+\sin 3\left(\frac{\pi}{6}\right)=1.5$	dM1A1 [4]
	$\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ 1.5\right)\right]$	
	ALT	
	Max of sine curve is 1 so that $\frac{1}{2}+1=\frac{3}{2} \Rightarrow y=\frac{3}{2}$	[M1A1
	$\sin 3\theta = 1 \ \Rightarrow\ 3\theta=\frac{\pi}{2} \Rightarrow\ \theta=\frac{\pi}{6}$	dM1A1]
	$\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ \frac{3}{2}\right)\right]$	
(c)	Uses the given x coordinate for point M	
	At point M $\quad\frac{dy}{dx}=3\cos\left(3\times\frac{7\pi}{18}\right)=-\frac{3\sqrt{3}}{2}$	B1
	$y-0=-\frac{3\sqrt{3}}{2}\left(x-\frac{7\pi}{18}\right) \Rightarrow 12y+18\sqrt{3}x-7\sqrt{3}\pi=0$ o.e.	M1A1A1 [4]
(d)	$A=\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2}+\sin 3x\right)dx+\left\lvert\left[\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2}+\sin 3x\right)dx\right]\right\rvert$	M1
	$A=\left[\frac{1}{2}x-\frac{\cos 3x}{3}\right]_0^{\frac{7\pi}{18}}+\left\lvert\left[\frac{1}{2}x-\frac{\cos 3x}{3}\right]_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\right\rvert$	M1
	$A=\left[\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{\cos 3\left(\frac{7\pi}{18}\right)}{3}\right)-\left(0-\frac{\cos 3\times 0}{3}\right)\right]$	M1

Type MS · Finding the equation of a tangent

Answer 8 of 9 · Question 10 part c · 4 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

	$+\left\|\left[\left(\frac{1}{2}\times\frac{11\pi}{18}-\frac{cos\,3\left('\frac{11\pi}{18}'\right)}{3}\right)-\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$ $A=[(1.23287)+(0.22828)]=1.46115\ldots\approx 1.46$	A1 [4]
		Total 15 marks

Type MS · Finding the equation of a tangent

Answer 8 of 9 · Question 10 part c · 4 marks

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	A1	For correctly obtaining $x = \frac{\pi}{6}$
(c)	**B1**	For correct gradient at point M $\left(\frac{dy}{dx}\right) = -\frac{3\sqrt{3}}{2}$
	M1	For a fully correct method of finding the equation of the tangent to C at M $y - 0 = ' -\frac{3\sqrt{3}}{2} ' \left(x - \frac{7\pi}{18}\right)$ Allow use of their $-\frac{3\sqrt{3}}{2}$ obtained from substitution of $x = \frac{7\pi}{18}$ into their $\frac{dy}{dx}$.
	A1	For a correct equation of the tangent in any form: $y - 0 = -\frac{3\sqrt{3}}{2}\left(x - \frac{7\pi}{18}\right)$
	A1	For correct equation of the tangent in the required form: $12y + 18\sqrt{3}x - 7\sqrt{3}\pi = 0$ Accept integer multiples of this equation.
(d)	**M1**	For identifying the correct limits to find the area above the curve and the area below the curve: $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) + \left|\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)\right|$ or $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) - \int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)$ Allow use of their $\frac{11\pi}{18}$. Needs to correctly indicate dealing with areas above and below axis.
	M1	For an attempt to integrate $\frac{1}{2} + \sin 3x$ obtaining: $\frac{1}{2}x - k\cos 3x$ where $k \neq -3$ For this mark ignore incorrect / absent limits.
	M1	For substituting limits correctly into their integrated expression (must be a changed expression). $A = \left[\left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right) - \left(0 - \frac{cos\,3 \times 0}{3}\right)\right]$ $+ \left\|\left[\left(\frac{1}{2} \times \frac{11\pi}{18} - \frac{cos\,3\left(\frac{'11\pi'}{18}\right)}{3}\right) - \left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$ Allow use of their $\frac{11\pi}{18}$. Must show substitution or M0.
	A1	For correctly obtaining awrt 1.46

Type MS · Finding the equation of a tangent

Answer 8 of 9 · Question 10 part c · 4 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding the equation of a tangent

Answer 9 of 9 · Question 6 part b · 5 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks
6(a)	$\frac{dy}{dx} = \frac{\left(x^2+1\right)2xe^{\left(x^2+1\right)} - 2xe^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$	M1A1A1
	$\frac{dy}{dx} = \frac{2xe^{\left(x^2+1\right)}\left(x^2+1-1\right)}{\left(x^2+1\right)^2} = \frac{2x^3e^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$	M1A1 cso [5]
(b)	When $x=-1$ $\frac{dy}{dx} = \frac{-e^2}{2}$, $y = \frac{e^2}{2}$	B1ft, B1
	$y - \frac{e^2}{2} = -\frac{e^2}{2}(x+1), \Rightarrow y = -\frac{e^2x}{2}$ oe	M1A1ft, A1 [5]
		Total 10 marks

Question	Notes	Marks
6(a)	$y = \frac{e^{\left(x^2+1\right)}}{x^2+1}$	
	Using Quotient Rule $\frac{dy}{dx} = \frac{\left(x^2+1\right)2xe^{\left(x^2+1\right)} - 2xe^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$ • For an attempt to differentiate both $e^{\left(x^2+1\right)}$ and x^2+1 Award for either $e^{x^2+1} \Rightarrow 2xe^{x^2+1}$ or $x^2+1 \Rightarrow 2x$ but both must be **changed** expressions. • Numerator is to have two terms in either order **subtracted.** • Denominator must be $\left(x^2+1\right)^2$	M1
	At least one term fully correct in the numerator	A1
	Fully correct unsimplified.	A1
	For an attempt to take out a common factor of either $2x$ or $e^{\left(x^2+1\right)}$ $\frac{dy}{dx} = \frac{2xe^{\left(x^2+1\right)}\left(x^2+1-1\right)}{\left(x^2+1\right)^2}$ OR Multiplies out the first term in the numerator $\frac{2x^3e^{\left(x^2+1\right)} + 2xe^{\left(x^2+1\right)} - 2xe^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$	M1 [M1]

Type MS · Finding the equation of a tangent

Answer 9 of 9 · Question 6 part b · 5 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

	Using Product Rule	
	$\frac{dy}{dx}=\left(x^2+1\right)^{-1}\times 2xe^{x^2+1}+(-2x)e^{x^2+1}\times\left(x^2+1\right)^{-2}$ • For an attempt to differentiate both $e^{\left(x^2+1\right)}$ and $\left(x^2+1\right)^{-1}$ Award for either $e^{x^2+1}\Rightarrow 2xe^{x^2+1}$ or $\left(x^2+1\right)^{-1}\Rightarrow(-2x)\left(x^2+1\right)^{-2}$ but both must be **changed** expressions. • Numerator is to have two terms added	M1
	At least one term correct	A1
	Fully correct unsimplified or simplified	A1
	For an attempt to take out a common factor of either $2x$ or $e^{\left(x^2+1\right)}$ and set a common denominator of $\left(x^2+1\right)^2$	M1
	For the fully correct expression for the derivative, $\frac{dy}{dx}=\frac{2x^3e^{\left(x^2+1\right)}}{\left(x^2+1\right)^2}$ NB: Any further simplification [e.g., cancelling $\left(x^2+1\right)$] following a correct answer seen, is A0.	A1 [5]
(b)	When $x=-1$, $\frac{dy}{dx}=\frac{-e^2}{2}$ ft their K in their $\frac{dy}{dx}$ Allow awrt $\frac{dy}{dx}=-3.7$ Even allow $\frac{dy}{dx}=\frac{-Ke^2}{4}$	B1ft
	When $x=-1$, $y=\frac{e^2}{2}$ Allow awrt $y=3.7$	B1
	For a correctly used method for the equation of the tangent using their values for $\frac{dy}{dx}$ and y $y-'\frac{e^2}{2}'='-\frac{e^2}{2}'(x+1)$ Also allow: $y-3.7=-3.7(x+1)$ If they use $y=mx+c$ they must obtain a value for c for the award of this mark.	M1
	For the correct equation in any form	A1ft
	For a correct simplified equation in any form but this must be in exact form. $y=-\frac{e^2x}{2}$ or $2y+e^2x=0$	A1 [5]
		Total 10 marks

Type MS · Finding the equation of a tangent

Answer 9 of 9 · Question 6 part b · 5 marks

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Question	Scheme	Marks